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#include <DHT.h>

#define DHTPIN 27
#define DHTTYPE DHT22
#define PIRPIN 5

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();
  pinMode(PIRPIN, INPUT);
  Serial.println("Smart Lab IoT - Simulation
  Started");
  delay(2000);
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  int motion = digitalRead(PIRPIN);
  if (isnan(temp) || isnan(hum)) {
```

```
temp = 24.5 + random(-10, 10) / 10.0;
hum = 60.0 + random(-5, 5);
}
Serial.print("Temp: "); Serial.print(temp);
Serial.print("C | Hum: "); Serial.print(hum);
Serial.print("% | Motion: ");
Serial.println(motion ? "Detected - LED ON" : "No
Motion");
delay(2000);
}
```